

Alphabet Inc Class C

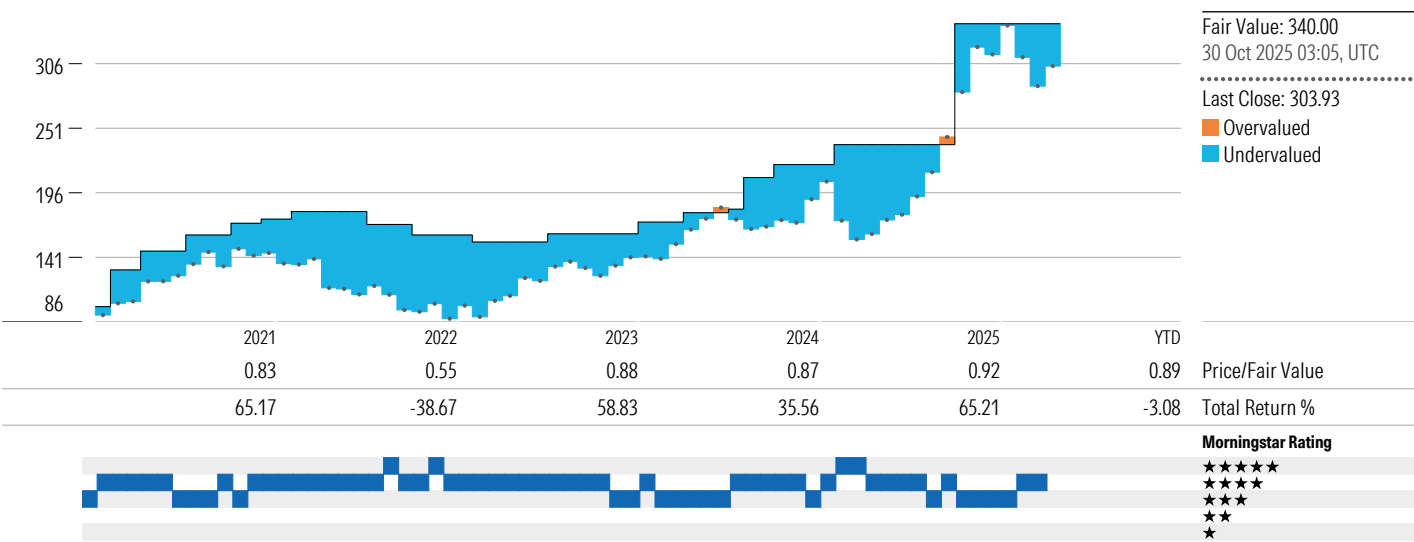
GOOG

★★★★★

7 Apr 2026 21:41, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
303.93 USD 7 Apr 2026	340.00 USD 30 Oct 2025 08:05, UTC	0.89	3.68 USD Tril 7 Apr 2026	Wide	Large Blend	Medium	Exemplary	 4 Mar 2026 06:00, UTC

Price vs. Fair Value



Total Return % as of 07 Apr 2026. Last Close as of 07 Apr 2026. Fair Value as of 30 Oct 2025 03:05, UTC.

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The primary analyst covering this company does not own its stock.

¹The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

Alphabet Is an Interest Behemoth With Its Fingers in Many Profitable Pies

Business Strategy & Outlook Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026

We view Alphabet as a conglomerate of stellar businesses. With solutions ranging from advertising to cloud computing and self-driving cars, Alphabet has built itself into a true behemoth, generating tens of billions of dollars in free cash flow annually. While antitrust concerns around Alphabet's core search business have made headlines, we retain our confidence in Alphabet's overall strength and foresee the firm remaining at the forefront of a variety of verticals, including search, artificial intelligence, video, and cloud computing.

Alphabet's core strategy is to preserve its strong advertising business, with the majority of advertising revenue coming from Google Search. To that end, the firm has invested considerably over the years to improve its search capabilities, ensuring that its search engine remains deeply embedded in how hundreds of millions of users access information on the web.

We see the firm's investments in AI as a continuation of this effort to safeguard its core product, Google Search. We believe that by leveraging generative AI, Google can not only improve its own search quality via features such as AI overviews, but also improve its advertising business by augmenting its ability to target customers with relevant ads.

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Sector

Communication Services

Industry

Internet Content & Information

Business Description

Alphabet is a holding company that wholly owns internet giant Google. The California-based company derives slightly less than 90% of its revenue from Google services, the vast majority of which is advertising sales. Alongside online ads, Google services houses sales stemming from Google's subscription services (YouTube TV and YouTube Music, among others), platforms (sales and in-app purchases on Play Store), and devices (Chromebooks, Pixel smartphones, and smart home products such as Chromecast). Google's cloud computing platform accounts for roughly 10% of Alphabet's revenue. The firm's investments in up-and-coming technologies such as self-driving cars (Waymo), health (Verily), and internet access (Google Fiber) make up the rest.

On the antitrust front, we don't foresee a material deterioration in Google's search business resulting from governmental or judicial intervention. While there is a range of possible outcomes depending on what remedial steps are imposed, we think it is likely that Google will maintain its leadership position in search and text-based advertising in the long term.

Beyond search, we have a positive outlook on Alphabet's cloud computing platform, Google Cloud Platform. We believe increased migration of workloads to the public cloud and an uptick in the deployment and usage of AI are key growth drivers for GCP over the next five years. At the same time, we believe that as GCP scales, it will become a more important part of Alphabet's overall business, both from a top-line and profitability perspective.

Bulls Say **Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026**

- ▶ Alphabet's core advertising business is deeply entrenched in advertising budgets, allowing the firm to benefit from a secular increase in digital advertising spending
- ▶ The firm's advertising business generates substantial cash flows that it can reinvest in growth areas such as GCP, AI-infused search, and aspirational projects such as Waymo.
- ▶ Alphabet has a huge opportunity in the lucrative public cloud space as a key cloud vendor to enterprises looking to digitize their workloads.

Bears Say **Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026**

- ▶ While Alphabet is seeking to diversify its business away from search, text-based advertising remains the largest contributor to the firm's top line, creating a concentration risk.
- ▶ Alphabet's continued investments in new, often unproven technologies have been a drag on cash flows.
- ▶ Regulators around the world are keying in on Alphabet's search dominance and could upend the search market through the imposition of deep, structural changes in the space.

Economic Moat **Malik Ahmed Khan, CFA, Senior Equity Analyst, 24 Oct 2025**

We believe Alphabet merits a wide economic moat rating owing to the intangible assets, network effect, cost advantage, and customer switching costs that permeate a variety of businesses that it owns.

While Alphabet's own reporting operating segments are split into Google services, Google Cloud, and other bets, we believe that for the purposes of analyzing the firm's economic moat and durable competitive advantage, a different split is more appropriate. In our moat analysis, we look at Google Search, YouTube, Google Cloud, Android and Google Play, devices, and other bets (which includes Google's aspirational projects such as self-driving vehicles and internet access).

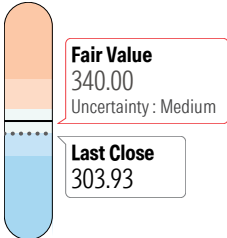
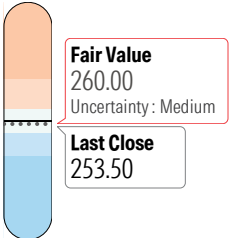
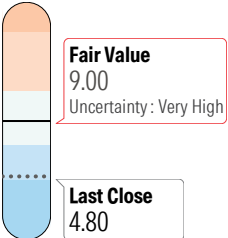
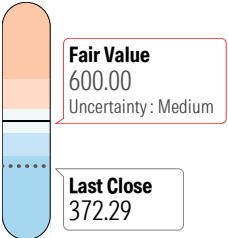
Google Search

We assign a wide moat rating to Alphabet's premier offering, Google Search. This general search engine has built significant intangible assets, primarily via its brand, and a potent network effect that allows

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Competitors

	Alphabet Inc Class C GOOG	Apple Inc AAPL	Snap Inc Class A SNAP	Microsoft Corp MSFT
				
Economic Moat	Wide	Wide	None	Wide
Currency	USD	USD	USD	USD
Fair Value	340.00 30 Oct 2025 03:05, UTC	260.00 30 Jan 2026 04:53, UTC	9.00 6 Aug 2025 04:01, UTC	600.00 31 Jul 2025 03:51, UTC
1-Star Price	459.00	351.00	15.75	810.00
5-Star Price	238.00	182.00	4.50	420.00
Assessment	Undervalued 7 Apr 2026	Fairly Valued 7 Apr 2026	Undervalued 7 Apr 2026	Undervalued 7 Apr 2026
Morningstar Rating	★★★★★ 7 Apr 2026 21:41, UTC	★★★★ 7 Apr 2026 21:26, UTC	★★★★★ 7 Apr 2026 21:43, UTC	★★★★★ 7 Apr 2026 21:23, UTC
Analyst	Malik Ahmed Khan, Senior Equity Analyst	William Kerwin, Senior Equity Analyst	Malik Ahmed Khan, Senior Equity Analyst	Dan Romanoff, Senior Equity Analyst
Capital Allocation	Exemplary	Exemplary	Standard	Exemplary
Price/Fair Value	0.89	0.98	0.53	0.62
Price/Sales	9.04	8.87	1.35	9.11
Price/Book	8.66	43.16	3.53	7.09
Price/Earning	32.56	32.77	—	23.61
Dividend Yield	0.29%	0.41%	0.00%	0.93%
Market Cap	3,676.64 Bil	3,721.67 Bil	8.11 Bil	2,764.49 Bil
52-Week Range	142.66—350.15	169.21—288.62	3.81—10.41	344.79—555.45
Investment Style	Large Blend	Large Blend	Small Blend	Large Blend

Alphabet to be the dominant player in the GSE space, with an engaged audience of both users and advertisers.

Google Search's dominance cannot be overstated. The firm's search engine has garnered and maintained ubiquity in the internet era to the degree that Merriam-Webster recognizes "google" as a verb that is interchangeable with search. Over the last two decades, Google Search has cemented its place as the most advanced GSE on the market, controlling more than 80% of the general search market since 2009. Market share shoots past 90% when looking at only mobile devices.

Google Search's intangible assets are related to the firm's brand, the general recognition of its GSE as the most advanced out there, and technological expertise when it comes to search algorithms, pricing mechanisms, and gathering valuable data for its advertising client base. Google Search's intangible assets also bolster its network effect. As more users engage with Google Search on account of its

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search superiority, it can monetize those users by selling better ads, which then allows the firm to invest more in Google Search to improve its GSE, while also collecting better signals for its advertising clients.

Google Search simultaneously solves two key problems for its main stakeholders, users and advertisers. A user wants access to high-quality information in a fast, seamless manner. Google provides this by keeping its products free and by constantly adding more functionality to its offering, such as live sports scores, shopping and travel tabs, and multimodal search options, among other features. Google collects various signals a user provides, such as clicks, time spent on a page, and time spent hovering over a link, to inform its advertising business, which sells real estate on the Google Search results page to advertisers.

Advertisers, on the other hand, want users to purchase their products/services. They have a finite amount of dollars to drive these sales and care deeply about the return on ad spending, or ROAS. When a user searches for a specific keyword, such as “QLED Television,” on Google Search, there is a high likelihood that the user is interested in buying a TV. In advertising nomenclature, by searching for a specific keyword, a user has signaled their intent. By getting ads in front of this hypothetical user, advertisers can convert the intent of the user into action, the purchase of a TV. As a result, Google Search is a core part of almost every digital advertising campaign, underscoring the high ROAS advertisers are able to generate via Google Search.

Alphabet has successfully monetized its Google Search offering, with the firm generating tens of billions of dollars in profits annually. We believe the firm's monetization also betrays a true sense of pricing power that stems from its intangible assets and network effect. By using a modified version of a second-price auction model, Google Search conducts auctions of the real estate on its main display page, with advertisers bidding on specific keywords. The auction itself is conducted in the milliseconds between a user search and the resulting display of results. Google has leveraged a series of “intentional pricing” maneuvers to increase bids and generate additional sales from its search engine. With its main competitor, Microsoft Bing, struggling to attract a critical mass of users, Google exercises significant pricing power in the general search market.

Google Search's dominance within the search market is also bolstered by a series of exclusive contracts Alphabet has signed with device manufacturers, such as Apple and Samsung, and third-party browsers, such as Mozilla Firefox. While these exclusive contracts are subject to increased regulatory scrutiny, they continue to channel significant traffic to Google Search, with approximately 50% of all general search inquiries in the United States stemming from a search access point covered by an exclusive contract. Alphabet, in return, offers a portion of its advertising revenue generated by these search points via a revenue-sharing agreement. While antitrust pressures may threaten Google Search's hold on the general search market, we don't believe there exists a scenario where an antitrust remedy

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pushes Google Search's returns on invested capital below its cost of capital.

When thinking of Google Search's placement as the default search engine for Apple and Android devices, it is important to delineate between the impact of default settings and Google Search's own product capabilities that bring users to its GSE by choice. One helpful illustration of the latter can be found in the PC market. While the majority of PCs in the world carry the Windows operating system, which has Microsoft Edge preloaded, the vast majority of Windows users opt to install Google Chrome and leverage Google Search instead of using Bing on Microsoft Edge.

Other competitive threats to Google Search come from specialized vendors such as Amazon, Meta, TikTok, and Yelp. While Alphabet certainly competes with all of the above in various verticals such as retail and travel, Google Search's dominance in navigational queries (queries to go to a specific website) and the general search market is unfazed by these specialized vendors. We also see them as competing in a slightly different market, with no reasonable user viewing Google Search and Meta as substitutes, for example. Similarly, advertisers view specialized vendors such as Amazon and Meta as complementary to ad spending on Google Search, with text-based ad budgets often controlled by different teams than Amazon/Meta ad placements.

Also, while generative search engines, such as SearchGPT by OpenAI, threaten to upend the general search market, we don't see those competitive threats resulting in a material deterioration of Google Search's excess returns. Irrespective of whether the new entrant is a well-capitalized player such as Apple or a venture-backed startup, the costs of building a GSE are cost-prohibitive. The new entrant has to build the infrastructure to support a GSE, which requires billions of dollars, while simultaneously attracting users and advertisers. After the initial infrastructure and customer acquisition, the entrant then needs to spend billions of dollars maintaining the GSE and investing in new capabilities such as better ad targeting, signal collection, and data storage. All of these costs, in sum, provide a structural barrier to competition for Google Search, which has scale that allows for cost-effective exposure to a large set of users and advertisers. Further, we believe Google's own generative AI offerings, such as Gemini, can avert any major customer/advertiser churn.

YouTube

We assign YouTube a wide moat rating owing to the business' strong network effect and intangible assets. Wrapped within Alphabet, YouTube is a streaming giant with an impressive brand, user base, and burgeoning advertising and subscription businesses.

YouTube, acquired by Alphabet in 2006, is one of the largest video-sharing/streaming platforms in the world. YouTube's scale is staggering. The platform has more than 2 billion monthly users with 1 billion hours of YouTube content viewed daily on TV alone. YouTube has parlayed its ubiquity to enter the subscription business as well. The platform's subscriptions, including YouTube Music and YouTube

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Premium, have more than 100 million subscribers, while YouTube TV, an offering that resembles the traditional TV bundle, boasts more than 8 million subscribers.

We believe YouTube's intangible assets, such as its large monetizable user base and brand, bolster its network effect. The platform's massive audience attracts creators seeking to monetize their videos, which, in turn, fuels more watch time. While more than 99% of YouTube's user base consists of free users, these users are critical for the platform's ad sales. By running a profitable advertising business, YouTube is able to invest in content, such as the seven-year deal to host the NFL Sunday Ticket, and boost its subscription offerings such as YouTube TV.

YouTube also offers Alphabet access to the top of the marketing funnel, allowing companies to create brand awareness that can spark a desire and intent to purchase a product. As a result, YouTube is a key part of Alphabet's overall advertising business as it is able to reach users in a medium very different than text-based search.

While the market landscape for YouTube is considerably more competitive than Google Search, we believe the firm's impressive ability to expand its reach across devices (phones, TVs, desktops) and its ability to innovate (creation of YouTube Shorts) should allow it to profitably run its platform and generate returns in excess of its cost of capital for the foreseeable future.

Google Cloud

While Alphabet's operating segments lump Google Workspace (productivity applications like Google Drive, Sheets, and Docs) together with Google Cloud Platform, we estimate that the vast majority of sales within Google Cloud stem from GCP, Alphabet's public cloud infrastructure arm. As a result, when investigating the competitive advantage for Google Cloud, we focus primarily on GCP.

We believe GCP merits a wide moat supported by high customer switching costs and cost advantages. While the Alphabet's public cloud platform is markedly smaller than Amazon's AWS or Microsoft's Azure, it is the third-largest public cloud vendor, controlling more than 10% of this lucrative market.

The cost advantages of GCP are obvious. The firm's continued investments in its cloud infrastructure, including AI, have enabled it to be one of three firms that have a global public cloud footprint. As compared with a customer building out their private cloud infrastructure, GCP, much like AWS and Azure, offers a significantly cheaper route that is analogous to renting real estate on GCP, helping the customer avoid massive capital expenditures associated with a private data center buildout. Further, as Alphabet has continued to invest in GCP, the economics of the business have improved dramatically. After generating substantial losses in prior years, GCP has turned profitable, with improving profitability on the horizon. As the firm expands its customer base, by typically offering better prices than AWS and Azure, it can continue to spread costs across its data center footprint, allowing its

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existing cost advantage to get stronger.

Another source of GCP's cost advantage has been Alphabet's investment in its own proprietary semiconductors, its TPU, or tensor processing unit. In lay terms, TPUs are custom-designed AI accelerators that can be leveraged for training and inference in the context of large AI models. While customers of GCP can opt to use third-party GPUs for their AI-related cloud workloads, all of Google's own AI development, such as the development of large language models, is done exclusively on its own TPUs. This represents a cost advantage over other peers using third-party GPUs to train their LLMs. While Microsoft and Amazon have been investing in their own AI chips, similar to Alphabet's TPUs, smaller public cloud vendors and customers choosing between public cloud or building their own data centers will likely remain cost-disadvantaged against GCP in this regard.

We also believe GCP benefits from high customer switching costs. Porting workloads from on-premises to the public cloud is not an easy lift, often requiring companies to rewire their IT infrastructure to allow for the transfer. This digital transformation is fraught with handover risk and cumbersome change management. Once a company, typically a large enterprise, has rolled into GCP, the costs of leaving GCP are significant. While data egress fees have been coming down throughout the public cloud space, the time and expense of application and database integrations have not. Similar to AWS and Azure, we think that the likelihood of customers changing mission-critical technology infrastructure is low. For context, when forecasting GCP's growth, we don't forecast AWS or Azure being customer donors to GCP, showcasing the high switching costs associated with public cloud vendors.

Android and Google Play

We believe Alphabet's Android operating system and Google Play, its application store present on all Android devices, enjoys a wide moat stemming from intangible assets and a network effect.

Android is the operating system for the majority of the world's smartphones, with more than two-thirds of all smartphones in the world running it. Virtually all smartphones, other than those manufactured by Apple, run on Android with the smartphone operating system space functionally a duopoly.

We believe Android and Google Play's intangible assets stem from the deals Alphabet has made with all smartphone manufacturers using Android as the default operating system. These deals, known as mobile application distribution agreements, allow smartphone manufacturers to use Alphabet's proprietary mobile applications that it has developed for the Android ecosystem. One of these proprietary mobile applications is Google Play, the app store that allows Android users to download non-preloaded applications such as Facebook or Spotify. In exchange for providing its applications, including Google Play, Alphabet gets prominent product placements for its applications, many of which are monetizable via advertising such as the Google Search widget, YouTube, and the Chrome internet browser.

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While these contracts are being challenged in courts as anticompetitive, final judgments remain far out in the future. We think that even if Alphabet is forced to open up its Android ecosystem by allowing smartphone manufacturers to leverage its applications on a different operating system, the successful entry of a new player in the operating system market is hard to envision.

The difficulty in creating an operating system is not simply tied to the expenses associated with engineering talent. The most crucial part of an operating system is its ability to connect its users to a wide range of applications beyond the ones that come preloaded with the device. This is where the network effect created by Google Play really shines.

Developers, from large companies and small, all know a basic truth: undertaking the costs of developing an application is only worth it if you can monetize the application by putting it in front of a large audience. The audience, in the case of smartphones, is almost exclusively controlled by Apple and Alphabet. This creates a virtuous cycle of developers working tirelessly to develop high-quality, bug-free applications for Android and iOS, which in turn attract more usage on these applications as users download them via Google Play and the App Store.

For a new entrant to enter the operating system market and compete with Android and Google Play, the entrant would have to simultaneously attract a critical mass of developers and users, neither wishing to enter a space that is not already populated by the other.

From a return on invested capital perspective, Android and Google Play are lucrative businesses. For every sale on Google Play, Alphabet collects a fee from the developer. This fee, after accounting for the cost of running Google Play, is likely incredibly lucrative from an operating margin perspective, indicating positive excess returns on the business unit. Similarly, the mobile application distribution agreements that Alphabet currently has with smartphone manufacturers include prominent placement of its own applications, serving as a funnel to its high-margin advertising business, which implies a high return on invested capital when it comes to operating and keeping Android up to date.

Devices

We believe that Alphabet's hardware portfolio, which includes the Pixel and Nest group of products, Chromecast, and Fitbit, does not merit an economic moat as a stand-alone business. This is primarily due to the lack of any material disclosure that would allow us to ascertain the business' profitability and returns generated.

While Pixel smartphone sales have shown strength over the last few years, including a surprising pocket of strength in Japan, where the smartphone constitutes roughly 10% of the overall smartphone market, we don't have a reliable method to determine the returns on the firm's invested capital in this

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space. We won't be surprised if Alphabet's hardware portfolio is generating excess returns and can see an argument for the presence of an economic moat, not too dissimilar to those built by other smartphone manufacturers, yet the obfuscation of granular data, an operational hazard when it comes to conglomerates such as Alphabet, precludes us from building a robust argument for an economic moat.

Other Bets

While Alphabet's other bets portfolio includes some interesting companies, including Waymo, which is one of the leaders in autonomous vehicles, the segment continues to burn capital, generating returns well below its cost of capital. As a result, we ascribe a no-moat rating to Alphabet's other bets business.

We believe that as some of the technologies Alphabet has invested in mature, they could spin off to be stand-alone businesses or perhaps be a larger part of the Alphabet story. We have particularly high hopes for Waymo, which has consistently shown promising results and progress in the AV market. We also understand that Alphabet, as a large incumbent, continues to invest in aspirational technologies such as Waymo as a way to be at the forefront of technological innovation. At the same time, however, there is considerable uncertainty as to whether any of Alphabet's aspirational investments will be economically successful.

Fair Value and Profit Drivers Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026

Our fair value estimate is \$340 per share, implying a 2026 adjusted price/earnings multiple of 31 times and an enterprise value/adjusted EBITDA multiple of 22 times.

We forecast Alphabet's top line growing at a 15% compound annual growth rate over the next five years.

Drilling deeper into the firm's various segments, we expect Google Search to grow at a high-single-digit level over the next five years as the digital advertising market matures and growth rates taper off after a robust few years following the pandemic. We expect YouTube to grow at a low-double-digit rate over the next five years, with a strong advertising business being increasingly supported by a robust subscription business.

We view Google Cloud as a key growth driver for Alphabet's overall business. We project Google Cloud sales to grow 35% annually on average over the next five years as cloud migrations, increased usage of AI, and additional software add-ons all work together to bolster the firm's cloud sales. Alongside the firm's public cloud business, we believe Google Cloud will also benefit from increased usage of Google's Workspace productivity applications as the firm embeds more AI tools within them, improving their quality while attracting higher average revenue per user. Over the next five years, we expect Google

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Cloud to constitute around 33% of Alphabet's overall top line, up from 15% at the end of 2025.

We foresee slight contraction in Alphabet's gross or operating margins over the next five years. While the firm's scale and improving Google Cloud margin profile should enable it to improve profitability, these potential improvements are more than offset by higher depreciation costs stemming from massive capital expenditures on data centers and AI capabilities.

As we model the trajectory of Alphabet's future revenue growth/profitability, we also look at hypothetical bear and bull cases.

The bear case includes a material deterioration in the firm's search revenue stemming from AI-led disruption. We model Google's search sales declining at a 4% annual rate over the next five years in our bear case as the firm loses search volume and market share to rivals such as OpenAI's Atlas and Perplexity's Comet. Some of the fair value impact implied by a decline in Google Search sales is offset by a decrease in traffic acquisition costs. Our fair value estimate in this bear case is \$230 per share.

The bull case envisions a scenario where Google Search maintains its dominance in the online search market while the video and cloud business continues to fire on all cylinders as AI adoption accrues material benefits to the firm with little AI-led disruption. Our fair value estimate in the bull case is \$400 per share.

We also look at valuing Alphabet using a sum-of-the-parts approach. Using SOTP, we arrive at a \$360 fair value estimate. While the majority of Alphabet's enterprise value continues to be driven by advertising, with Google Search being the largest advertising segment, we see a lot of potential value in other areas of Alphabet as well. According to our SOTP valuation, Google Cloud constitutes nearly 31% of the firm's overall intrinsic value. YouTube ex ads, Play, and hardware sales along with Waymo combined make up 16% of Alphabet's total value.

Risk and Uncertainty Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026

We assign Alphabet a Morningstar Uncertainty Rating of Medium. This reflects our belief that despite the near-term uncertainty around antitrust regulation and potential competition in the AI-infused search market, Alphabet is well positioned to expand its overall business while maintaining a rock-solid balance sheet.

We believe Google's intangible assets and network effects will likely safeguard its dominance in the search space. Further, the firm's continued investments in AI, which Alphabet can leverage across nearly every business it operates, should be value-accretive. We do think Google Search's status as the runaway leader in search could come under pressure, primarily due to the antitrust scrutiny. While we don't see the firm's market leadership slipping due to antitrust concerns, this issue adds uncertainty to an otherwise stable business.

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Alongside antitrust, the other key area of uncertainty for Alphabet as a whole is AI. The firm has invested, and continues to invest, a considerable amount of capital in new AI technologies such as generative AI. While we view these investments as value-accretive, especially considering the myriad ways Alphabet can monetize its technological know-how in AI, we view them as adding more uncertainty to the range of possible outcomes for Alphabet's top line and profitability in the future.

The firm's high dependence on user behavior data represents an environmental, social, and governance risk. If it fails to maintain adequate data privacy and security, Google's advertising business would likely suffer, and users' trust in the firm's other products would likely falter.

Capital Allocation Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026

We assign Alphabet an Exemplary Capital Allocation Rating based on its sound balance sheet, exceptional investments, and appropriate shareholder distributions.

Alphabet's financial health is enviable. With the company's more than \$127 billion in cash and cash equivalents, we are not concerned about the \$46 billion in debt on the balance sheet. Alongside incredibly low leverage ratios, Alphabet's cash generation profile remains stellar.

Alphabet's investment strategy has also proved to be shrewd and value-accretive over the past two decades. Alongside smart M&A decisions, which include the acquisitions of Android, YouTube, DoubleClick, Fitbit, and Mandiant, Alphabet has aggressively invested in research and development over the years. Its research spending as a proportion of total sales has remained in the 14%-16% range over the last seven years, a trend we see continuing as the firm invests in nascent technologies such as generative AI.

With combative antitrust regulators scrutinizing Alphabet's every move, we think it is unlikely that Alphabet will be able to engage in any blockbuster M&A in the near future.

Alphabet's shareholder distributions have also been solid. The firm has a massive share-buyback program that has enabled it to reduce its shares outstanding by 9% over the last five years. Further, Alphabet pays a quarterly dividend of \$0.21 on top of regular share repurchases. We expect the firm to leverage both dividend and share repurchases as avenues of shareholder distributions in the future.

Larry Page and Sergey Brin co-founded Google. In December 2019, Page and Brin left their roles as Alphabet CEO and president, respectively, but remain on the board. Sundar Pichai, the CEO of Google, also became the CEO of Alphabet. Pichai joined Google in 2004 and was its product chief before becoming CEO in 2015. Although management's decisions have generated exceptional returns for shareholders in the past, we remain watchful regarding the high concentration of voting power. Page and Brin hold more than 50% voting power through Class B supervoting shares.

Analyst Notes Archive

Alphabet Inc Class C **GOOG** ★★★★★ 7 Apr 2026 21:41, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment ¹
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Alphabet Earnings: Continued Momentum on Cloud and Search as Capital Expenses Set to Mount

Malik Ahmed Khan, CFA, Senior Equity Analyst, 5 Feb 2026

Alphabet reported strong fourth-quarter earnings, with sales up 18% to \$113 billion and adjusted operating margins down 50 basis points to 31.6%. Google Cloud continues to be the star of the show, accelerating both sequentially and year over year to 48% growth. Why it matters: Long gone are the days when the market had written off Alphabet as an artificial intelligence laggard. The massive surge in Alphabet shares has closely mirrored a turning tide, with investors now viewing the firm as an AI winner, with AI driving sales across the firm's many segments. The most obvious beneficiary of AI within Alphabet remains its cloud business, which now constitutes 16% of the firm's total top line. The launch of Gemini 3, Alphabet's latest large language model, continued to expand the firm's enterprise and consumer market share. We are impressed at how Alphabet continues to integrate AI within Google Search. By adding features such as AI Overviews and AI Mode, the firm has not only mitigated a real competitive threat from GenAI chatbots but also increased the number of queries and the ad price per query. The bottom line: We maintain our \$340 fair value estimate for wide-moat Alphabet and continue to view the firm as a clear leader in AI. Shares traded slightly down after hours, and we view them as fairly valued. We continue to applaud Alphabet's full-stack AI strategy, which enables the firm to be the master of its own destiny. Unlike peers that rely on external AI labs and AI chips, Alphabet's investments in Google DeepMind and TPUs appear increasingly prescient over time. Big picture: While results remain upbeat, it is important to step back and remain cognizant of the costs associated with AI. Management guided to approximately \$180 billion in capital expenditures in 2026, a 97% year-over-year increase, and 38% of our sales forecast for the year. Coming up: We expect a broader rollout of ads within AI Search in 2026, which should add incremental Google Search sales.

Alphabet: Reports Indicate That Company Is Considering Ads Within Gemini in 2026

Malik Ahmed Khan, CFA, Senior Equity Analyst, 8 Dec 2025

Adweek reported that Alphabet has indicated to advertisers that the firm is working on rolling out ads on Gemini, its AI chatbot, in 2026. Alphabet's shares sold off more than 2% as investors question the product impact of ads and whether ads could negatively affect Gemini's strong momentum. Why it matters: We see this move, if properly executed, as value-additive for Alphabet. The firm's distribution on Gemini, which has more than 650 million monthly users, could grow materially if it offers more powerful artificial intelligence tools to more users in exchange for ads placed in Gemini. While according to reports OpenAI is deprioritizing its ads push to focus more on the product, Alphabet appears to be focusing on monetization. We reiterate our view that subscriptions are a sub-optimal monetization vector as the vast majority of global users simply do not wish to part with \$20 per month for AI tools. We believe that, like search, we will see ads emerge as the optimal monetization strategy for GenAI chatbots. The bottom line: We retain our \$340 fair value estimate for wide-moat Alphabet. If the

Alphabet Inc Class C **GOOG** ★★★★★ 7 Apr 2026 21:41, UTC

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firm launches ads and can provide advertisers a high return on ad spending, or ROAS, while growing its user base, we expect there to be upside to our valuation as more ad dollars would flow into Alphabet. Google Search has an average revenue per user, or ARPU, of \$44. Put another way, Alphabet sells ads worth \$44 per user. While subscription offers much more upside per paying user, the paid to total user ratio is incredibly low, with ChatGPT's ratio of paid users at roughly 5%. As a result, we think introducing ads would be Alphabet pursuing what we view to be the optimal strategy, one that includes both an unpaid, ad-supported tier and a subscription-based, no-ad tier. Big picture: We think Alphabet's expertise in advertising will drive Gemini's ROAS higher and make the platform an attractive proposition for advertisers looking for GenAI traffic to their products.

Alphabet: Strong Gemini Momentum and Possible Meta TPU Deal Underscore Alphabet's AI Chops

Malik Ahmed Khan, CFA, Senior Equity Analyst, 25 Nov 2025

Alphabet's recently launched Gemini 3 model and rumored TPU deal with Meta, which would potentially include TPUs being placed in Meta's data centers, have sent shares of the firm higher in the last week as the market repositions Alphabet as a standout artificial intelligence leader. Why it matters: We have reiterated our confidence in Alphabet's full-stack AI strategy multiple times over the last year, with the firm being the only AI player with leading-edge solutions all the way from infrastructure, TPUs, and Google Cloud, to consumer distribution, Search, and Gemini. We believe that Gemini 3's outperformance on not only performance benchmarks, which can be "gamed," but also in actual use as evidenced by the rave reviews it has generated from the developer community, demonstrate the firm's AI chops. We also view the rumored Meta TPU deal as significant as it would be the first time TPUs would be placed outside of Google Cloud, opening up another revenue stream for Alphabet if the firm scales external TPU sales and placing it in direct competition with the likes of Nvidia. The bottom line: We maintain our \$340 fair value estimate for wide-moat Alphabet and continue to view the firm as exceptionally well positioned in everything AI. With shares up almost 20% since the firm's third-quarter earnings at the end of October, we view them as fairly valued. While we remain optimistic about Alphabet's TPU program, we view it as primarily an outpost to drive more Google Cloud sales while expanding Alphabet's AI monetization surface. If external TPU sales ramp meaningfully, we see further upside to our fair value estimate. Big picture: We believe multimodality built into Alphabet's Gemini models continues to be a big driver of usage. We saw a surge in Gemini's monthly users after the launch of its image generation model, Nano Banana, in the summer and expect another surge following the successful Gemini 3 launch.

Alphabet Earnings: Let's Google; Search Strength and Cloud Momentum Drive our FVE to \$340 From \$300

Malik Ahmed Khan, CFA, Senior Equity Analyst, 30 Oct 2025

Alphabet reported solid third-quarter earnings with sales growing 16% to \$102 billion and adjusted operating margins expanding 160 basis points to 34%. Google Cloud continues to fire on all cylinders,

Alphabet Inc Class C **GOOG** ★★★★★ 7 Apr 2026 21:41, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
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accelerating sequentially to 34% growth in the quarter, constituting 15% of total sales. Why it matters: Alphabet's execution on artificial intelligence, evidenced by strong traction for its Gemini app, which has more than 650 million monthly users, along with its ability to deliver solid advertising revenue, continues to drive results while refuting the AI-led disruption narrative. Within search, we like AI Overviews and AI Mode as tools that not only retain users but also drive incremental query growth and improved user engagement. With higher query growth driven by these AI features, Alphabet can increase its ad inventory, lifting sales in the process. Also, as agent-driven web browsing takes off, having access to and control over Chrome's multibillion-user base will be crucial, as agentic commerce features could allow Alphabet to command a hefty take rate on agent-led commerce activity, unlocking a new growth vector. The bottom line: We are raising our fair value estimate for wide-moat Alphabet to \$340 from \$300 following the firm's quarterly results, which blew past our expectations. The 79% growth in the firm's Google Cloud backlog, up from 37% last quarter, points to an inflection in cloud demand coming in 2026. Based on the massive increase in Google Cloud's backlog, we are raising our Google Cloud five-year growth to 34%, up from 30% in our prior forecast. This increase is the key driver behind our fair value increase. In the past six months, Alphabet's stock has risen more than 75% as the market has come to agree with our view of Alphabet as an AI leader, not a laggard. With our fair value increase, we see further upside as Google Cloud continues to drive AI monetization while search remains solid.

Alphabet: Anthropic Deal and Gemini Momentum Stand to Deliver; Fair Value Up 27% Malik Ahmed Khan, CFA, Senior Equity Analyst, 24 Oct 2025

Anthropic announced a material expansion of its use of Alphabet's AI accelerators. The deal, worth tens of billions of dollars, includes the use of Alphabet's tensor processing units as well as traditional Google Cloud services, with more than 1 gigawatt of power coming online in 2026. Why it matters: We believe Anthropic's multibillion-dollar commitment is a vote of confidence in Alphabet's TPUs as well as a material revenue driver. As artificial intelligence labs hunt for capacity, we see TPUs only growing in importance. With 1 GW of capacity earmarked for Anthropic in 2026, we believe this will bring in around \$8 billion-\$10 billion in sales annually, driving substantial growth in Google Cloud sales. We also view Gemini's continued momentum, with the recent viral launch of Nano Banana, an image-focused model, as evidence of Alphabet flexing its AI muscles in the consumer space, a trend we expect to accelerate. The bottom line: We raise our fair value estimate for wide-moat Alphabet to \$300 per share from \$237 as we bake in faster cloud growth and improved longer-term profitability. We view the shares as modestly undervalued going into the third-quarter earnings release on Oct. 29. We believe that the Anthropic deal may push other frontier labs still leveraged to Nvidia's GPUs to rethink their training and inference silicon roadmaps and encourage them to include TPUs as part of their AI infrastructures, especially as capacity remains constrained. From a strategic perspective, we believe TPUs offer Alphabet not only a cost advantage as it comes to training and running inference on its own

Alphabet Inc Class C

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Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
303.93 USD	340.00 USD	0.89	3.68 USD Tril	Wide	Large Blend	Medium	Exemplary	4 Mar 2026 06:00, UTC
7 Apr 2026	30 Oct 2025 08:05, UTC		7 Apr 2026					

models but also an alternative monetization angle for its large AI capital expenditures.Coming up: We expect the earnings report to further showcase Alphabet's ability to not only benefit from continued AI monetization in Google Cloud but also retain double-digit growth in its core search business.

Alphabet: Remedies Proposed in the Google Search Case Do Not Pose a Material Risk to Alphabet

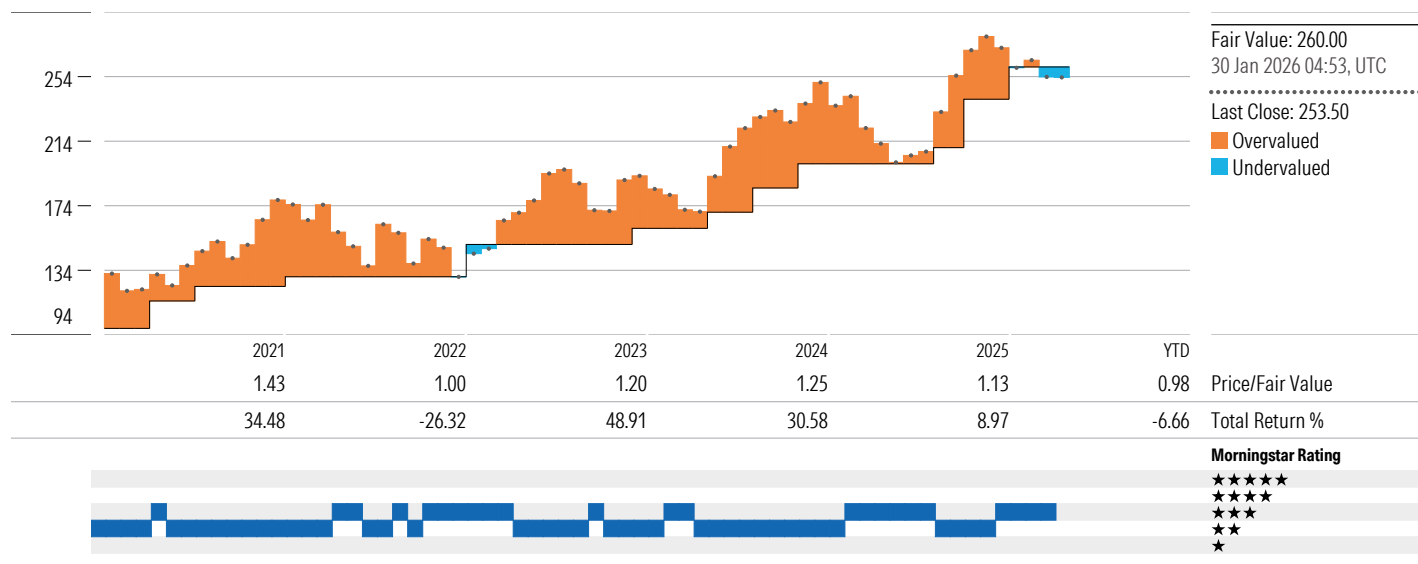
Malik Ahmed Khan, CFA,Senior Equity Analyst,3 Sep 2025

On Sept. 2, a federal judge published his proposed remedies in the Google Search antitrust case. Alphabet won't be required to divest Chrome or Android. The company will not be allowed to have exclusive search agreements and will need to engage in modest data sharing with competitors. Why it matters: The Google Search case has been hanging over Alphabet's head. With this ruling, investors can breathe a sigh of relief, knowing that US District Judge Amit Mehta did not propose structural remedies that could have been value-destructive. Our view, which we iterated in various notes published in August, October, and November of 2024, as well as in February, April, and July of 2025, was largely ratified by Mehta's decision. We had previously highlighted that the likely remedies would be behavioral in nature. At the same time, even the imposition of these remedies is not immediate, with an Alphabet appeal to this case likely elongating the process by a few years. This elongation will occur as the search market continues to evolve, likely leading to an overall dilution of the proposed remedies. The bottom line: We maintain our \$237 fair value estimate as our base-case outlook, as the antitrust opinion published today largely upholds our outlook. With shares jumping 7% following the decision, we now view Alphabet as fairly valued, with the stock up more than 50% from its April lows. We believe that with this crucial remedies opinion out of the way, Alphabet can focus more clearly on its search business while aggressively integrating artificial intelligence features as the search market goes through a period of transformation from traditional to AI-first search. ■■■

Alphabet Inc Class C **GOOG** ★★★★★ 7 Apr 2026 21:41, UTC

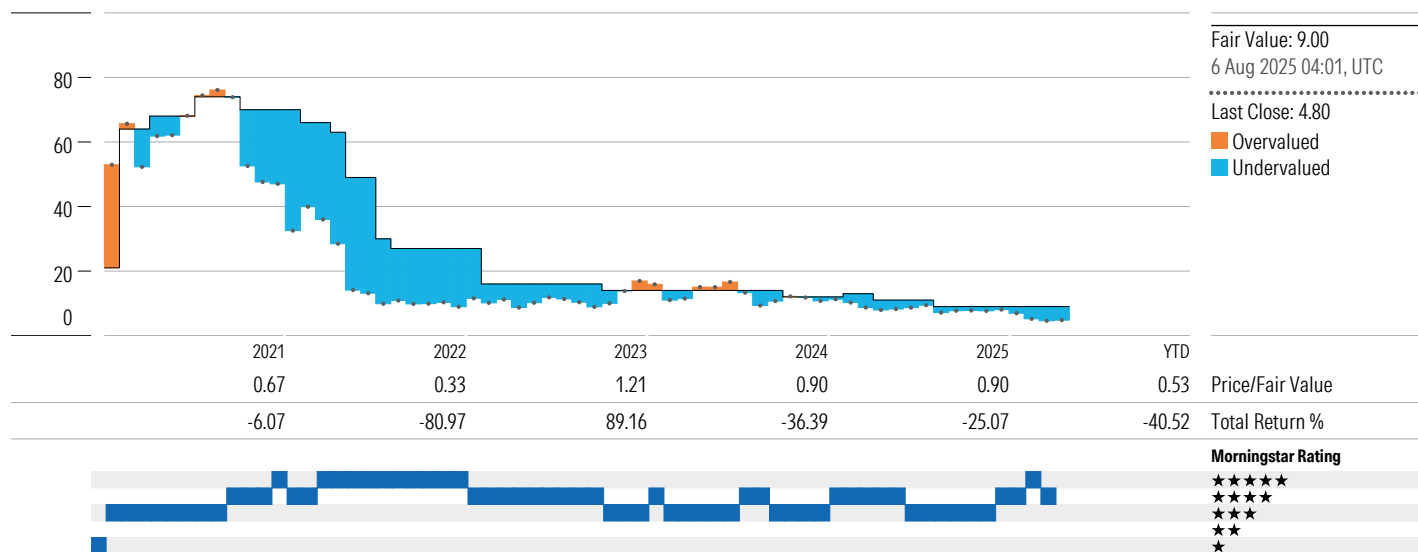
Competitors Price vs. Fair Value

Apple Inc AAPL



Total Return % as of 07 Apr 2026. Last Close as of 07 Apr 2026. Fair Value as of 30 Jan 2026 04:53, UTC.

Snap Inc Class A SNAP



Total Return % as of 07 Apr 2026. Last Close as of 07 Apr 2026. Fair Value as of 6 Aug 2025 04:01, UTC.

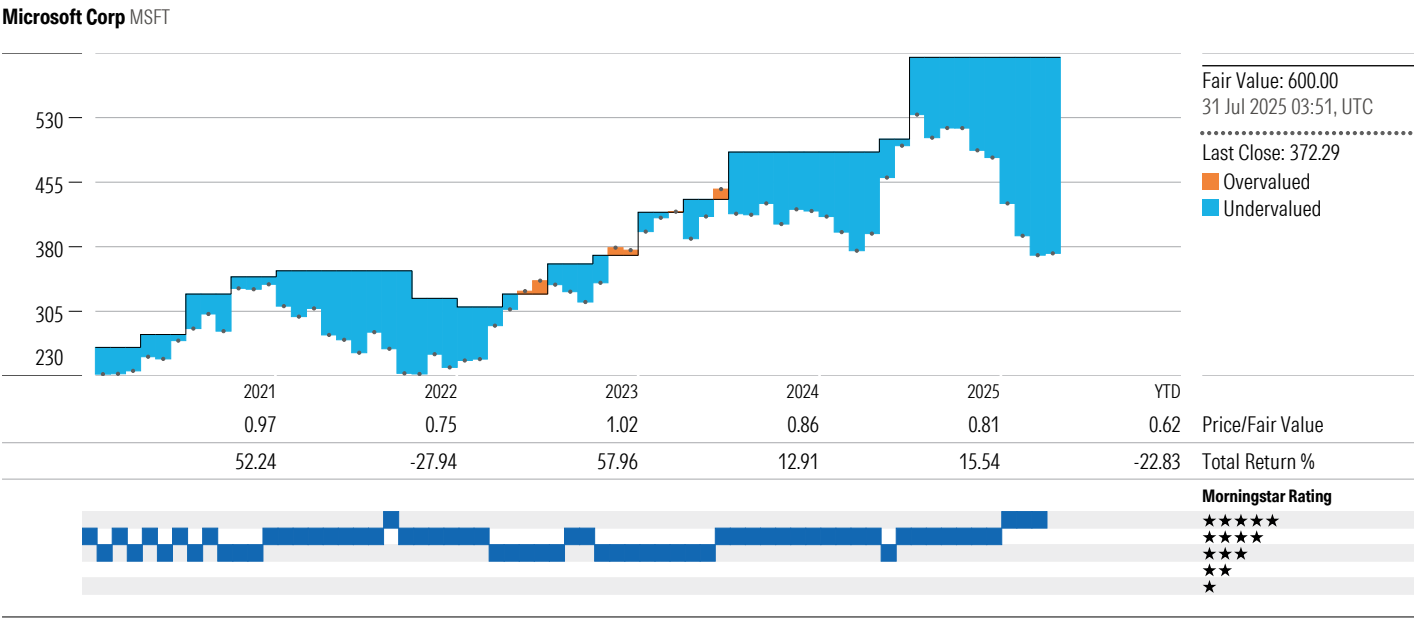
Alphabet Inc Class C

GOOG

★★★★★

7 Apr 2026 21:41, UTC

Competitors Price vs. Fair Value



Total Return % as of 07 Apr 2026. Last Close as of 07 Apr 2026. Fair Value as of 31 Jul 2025 03:51, UTC.

Alphabet Inc Class C GOOG ★★★★★ 7 Apr 2026 21:41, UTC

Last Price 303.93 USD 7 Apr 2026	Fair Value Estimate 340.00 USD 30 Oct 2025 08:05, UTC	Price/FVE 0.89	Market Cap 3.68 USD Tril 7 Apr 2026	Economic Moat™ Wide	Equity Style Box Large Blend	Uncertainty Medium	Capital Allocation Exemplary	ESG Risk Rating Assessment¹  4 Mar 2026 06:00, UTC
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Morningstar Valuation Model Summary

Financials as of 04 Feb 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (USD Mil)	307,394	350,018	402,836	472,401	542,847	618,918	704,277	799,812
Operating Income (USD Mil)	84,293	112,390	129,039	144,909	166,653	190,302	214,183	250,303
EBITDA (USD Mil)	96,239	127,701	150,175	178,449	210,081	236,721	264,186	307,089
Adjusted EBITDA (USD Mil)	96,239	127,701	150,175	178,449	210,081	236,721	264,186	307,089
Net Income (USD Mil)	73,795	100,118	132,170	133,995	152,477	172,579	192,877	223,580
Adjusted Net Income (USD Mil)	73,795	100,118	132,170	133,995	152,477	172,579	192,877	223,580
Free Cash Flow To The Firm (USD Mil)	54,702	53,627	37,961	-18,904	37,041	70,129	103,393	154,832
Weighted Average Diluted Shares Outstanding (Bil)	13	12	12	12	12	12	12	12
Earnings Per Share (Diluted) (USD)	5.80	8.04	10.81	10.91	12.53	14.29	16.07	18.73
Adjusted Earnings Per Share (Diluted) (USD)	5.80	8.04	10.81	10.91	12.53	14.29	16.07	18.73
Dividends Per Share (USD)	0.00	0.80	0.84	0.88	0.93	0.97	1.02	1.07

Margins & Returns as of 04 Feb 2026

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	30.5	27.4	32.1	32.0	30.7	30.7	30.8	30.4	31.3	30.8
EBITDA Margin %	—	31.3	36.5	37.3	37.8	38.7	38.3	37.5	38.4	—
Adjusted EBITDA Margin %	—	31.3	36.5	37.3	37.8	38.7	38.3	37.5	38.4	38.1
Net Margin %	28.5	24.0	28.6	32.8	28.4	28.1	27.9	27.4	28.0	28.0
Adjusted Net Margin %	28.5	24.0	28.6	32.8	28.4	28.1	27.9	27.4	28.0	27.9
Free Cash Flow To The Firm Margin %	14.2	17.8	15.3	9.4	-4.0	6.8	11.3	14.7	19.4	9.6

Growth & Ratios as of 04 Feb 2026

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	12.5	8.7	13.9	15.1	17.3	14.9	14.0	13.8	13.6	14.7
Operating Income Growth %	19.9	12.6	33.3	14.8	12.3	15.0	14.2	12.6	16.9	14.2
EBITDA Growth %	19.8	9.0	32.7	17.6	18.8	17.7	12.7	11.6	16.2	15.4
Adjusted EBITDA Growth %	19.4	9.0	32.7	17.6	18.8	17.7	12.7	11.6	16.2	15.4
Earnings Per Share Growth %	33.4	27.3	38.7	34.4	0.9	14.9	14.1	12.5	16.5	11.6
Adjusted Earnings Per Share Growth %	33.4	27.3	38.7	34.4	0.9	14.9	14.1	12.5	16.5	11.6

Valuation as of 04 Feb 2026

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earning	24.3	23.7	29.0	27.9	24.3	21.3	18.9	16.2
Price/Sales	5.7	6.6	9.4	7.8	6.8	6.0	5.2	4.6
Price/Book	6.3	7.3	9.2	7.5	6.2	5.0	4.1	3.4
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	17.3	17.6	24.8	20.3	17.2	15.3	13.7	11.8
EV/EBIT	19.7	20.0	28.9	25.0	21.7	19.0	16.9	14.5
Dividend Yield %	—	0.4	0.3	0.3	0.3	0.3	0.3	0.4
Dividend Payout %	0.0	10.0	7.8	8.1	7.4	6.8	6.4	5.7
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 04 Feb 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	18.3	22.2	22.2	19.2	18.6	17.8	16.9	16.5
ROE %	26.0	30.8	31.8	27.0	25.4	23.7	21.9	21.0
ROIC %	35.3	36.3	33.4	27.9	24.7	23.4	22.7	23.3

Alphabet Inc Class C **GOOG** ★★★★★ 7 Apr 2026 21:41, UTC

Last Price 303.93 USD 7 Apr 2026	Fair Value Estimate 340.00 USD 30 Oct 2025 08:05, UTC	Price/FVE 0.89	Market Cap 3.68 USD Tril 7 Apr 2026	Economic Moat™ Wide	Equity Style Box Large Blend	Uncertainty Medium	Capital Allocation Exemplary	ESG Risk Rating Assessment¹ 4 Mar 2026 06:00, UTC
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Financial Leverage (Reporting Currency)	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Year, ends 31 Dec								
Debt/Capital %	1.5	1.1	—	1.0	1.0	0.9	0.8	0.8
Assets/Equity	1.4	1.4	1.4	1.4	1.4	1.3	1.3	1.3
Net Debt/EBITDA	-0.9	-0.5	—	-0.1	-0.1	-0.3	-0.5	-0.9
Total Debt/EBITDA	0.3	0.2	—	0.3	0.2	0.2	0.2	0.2
EBITDA/ Net Interest Expense	-67.6	-17.2	-5.0	-14.0	-16.5	-18.6	-20.7	-24.1
Forecast Revisions as of 4 Feb 2026	2026		2027		2028			
Prior data as of 29 Oct 2025	Current	Prior	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	340.00	339.76	—	—	—	—	—	—
Revenue (USD Mil)	472,401	399,129	542,847	455,076	618,918	513,405		
Operating Income (USD Mil)	144,909	129,836	166,653	143,335	190,302	160,972		
EBITDA (USD Mil)	178,449	151,788	210,081	175,645	236,721	202,045		
Net Income (USD Mil)	133,995	131,239	152,477	138,211	172,579	153,202		
Earnings Per Share (Diluted) (USD)	10.91	10.75	12.53	11.59	14.29	13.13		
Adjusted Earnings Per Share (Diluted) (USD)	10.91	10.75	12.53	11.59	14.29	13.13		
Dividends Per Share (USD)	0.88	0.84	0.93	0.88	0.97	0.93		

Key Valuation Drivers as of 04 Feb 2026

Cost of Equity %	9.0
Pre-Tax Cost of Debt %	5.5
Weighted Average Cost of Capital %	9.0
Long-Run Tax Rate %	15.0
Stage II EBI Growth Rate %	10.5
Stage II Investment Rate %	25.6
Perpetuity Year	20

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 04 Feb 2026

	USD Mil
Present Value Stage I	201,124
Present Value Stage II	1,733,583
Present Value Stage III	2,110,804
Total Firm Value	4,045,510
Cash and Equivalents	126,843
Debt	46,547
Other Adjustments	0
Equity Value	4,125,806
Projected Diluted Shares	12
Fair Value per Share (USD)	340.00

Alphabet Inc Class C **GOOG** ★★★★★ 7 Apr 2026 21:41, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment ¹
303.93 USD 7 Apr 2026	340.00 USD 30 Oct 2025 08:05, UTC	0.89	3.68 USD Tril 7 Apr 2026	Wide	Large Blend	Medium	Exemplary	

ESG Risk Rating Breakdown

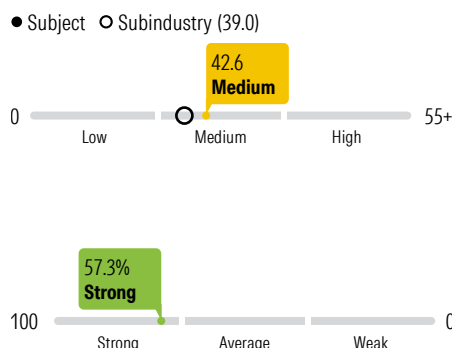
Exposure

Company Exposure ¹	42.6
- Manageable Risk	39.7
Unmanageable Risk²	2.9

Management

Manageable Risk	39.7
- Managed Risk ³	22.7
Management Gap⁴	17.0

Overall Unmanaged Risk 19.9



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 57.3% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment⁵



ESG Risk Rating is of Mar 04, 2026. Highest Controversy Level is as of Mar 08, 2026. Sustainalytics Subindustry: Internet Software and Services. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 04 Mar 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure		Management		ESG Risk Rating	
Alphabet Inc	42.6 Medium	0 ——— ● ——— 55+	57.3 Strong	100 ——— ● ——— 0	19.9 Low	0 ——— ● ——— 40+
Microsoft Corp	32.8 Low	0 ——— ● ——— 55+	60.3 Strong	100 ——— ● ——— 0	14.4 Low	0 ——— ● ——— 40+
Amazon.com Inc	34.7 Low	0 ——— ● ——— 55+	56.3 Strong	100 ——— ● ——— 0	16.6 Low	0 ——— ● ——— 40+
Apple Inc	33.7 Low	0 ——— ● ——— 55+	58.8 Strong	100 ——— ● ——— 0	15.3 Low	0 ——— ● ——— 40+
Snap Inc	38.2 Medium	0 ——— ● ——— 55+	51.4 Strong	100 ——— ● ——— 0	20.0 Low	0 ——— ● ——— 40+

Appendix

Historical Morningstar Rating

Alphabet Inc Class C GOOG 7 Apr 2026 21:41, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	★★★★	★★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★★	★★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★	★★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★

Apple Inc AAPL 7 Apr 2026 21:26, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	★★★	★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★	★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★	★★	★★	★★	★★	★★	★★	★★	★★★	★★★	★★	★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★	★★	★★★★	★★	★★	★★	★★	★★	★★★	★★★	★★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★★	★★	★★★★	★★	★★	★★★★	★★★	★★	★★	★★	★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★★	★★	★★	★★	★★	★★★	★★	★★	★★	★★

Snap Inc Class A SNAP 7 Apr 2026 21:43, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	★★★★	★★★★★	★★★★	★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★★★	★★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★	★★★★	★★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★

Microsoft Corp MSFT 7 Apr 2026 21:23, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★	★★★	★★★	★★★★	★★★	★★★★	★★★	★★★★	★★★	★★★★	★★★	★★★★

Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a

long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on these projections, we calculate earnings before interest,

after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



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outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

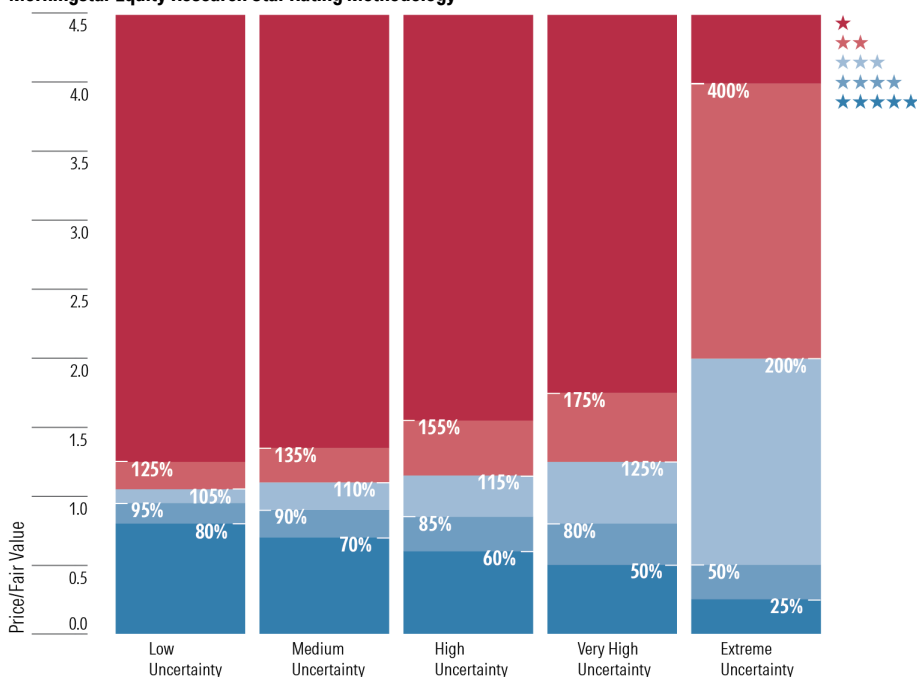
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

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Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score.

Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

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